

# Nicholas Muchelemba

**Location:** Lusaka, Lusaka Province, Zambia

**Email:** nmuchelemba60@gmail.com

**Phone:** 096 837 3024

**LinkedIn:** [linkedin.com/in/nicholas-muchelemba](https://www.linkedin.com/in/nicholas-muchelemba)

## Professional Summary

Backend and API engineer with 4+ years of experience building and scaling secure, cloud-based systems across healthcare, finance, and mining domains. Specializes in designing reliable RESTful APIs and backend services using Python (FastAPI), Java, and AWS. Known for improving system performance, reducing manual processes, and enabling data-driven decision-making. Certified AWS Solutions Architect and Certified Kubernetes Administrator.

## Professional Experience

### API Software Engineer — Orbis International

**Jul 2024 – Present | Remote**

- Led the design and development of scalable RESTful APIs powering global telemedicine and AI-enabled healthcare platforms
- Improved API reliability and performance through cloud-native practices and structured error handling
- Collaborated with architects and cross-functional teams to design secure, maintainable backend services aligned with platform architecture
- Built and maintained backend services using Python (FastAPI) and Java, enabling seamless integration with internal and third-party systems
- Supported deployment and operation of API infrastructure across multiple environments, ensuring high availability and security
- Integrated third-party services, reducing integration friction and improving system interoperability
- Contributed to backend architecture decisions, testing strategies, and deployment workflows to support continuous delivery and scalability

**Technologies:** Python (FastAPI), Java, REST APIs, AWS, Cloud Architecture

### Software Developer (Backend & Cloud)

**Feb 2023 – Jun 2024**

- Developed and maintained backend services and RESTful APIs using Python, Java, and FastAPI for cloud-based business applications
- Implemented data processing and automation solutions using AWS services (S3, Athena, Translate, Kendra, SageMaker), improving data accessibility and operational efficiency

- Designed and delivered Power BI dashboards to support analytics and decision-making
- Built backend integrations for internal and external systems, enhancing interoperability and reliability
- Contributed across the full software development lifecycle, including requirements analysis, development, testing, deployment, and support

**Technologies:** Python, Java, FastAPI, AWS (S3, Athena, Translate, Kendra, SageMaker), Power BI, Databases

## **Software Developer — First Quantum Minerals**

**Apr 2021 – Mar 2023**

- Developed and maintained internal business applications using C# (.NET/Blazor) to support mining operations and reporting workflows
- Designed and delivered Power BI reports and dashboards, improving visibility into operational and production data
- Built Python-based automation scripts to reduce manual processes and improve efficiency in data and reporting tasks
- Applied machine learning techniques to enhance prediction and analysis of mining BCM data
- Collaborated with cross-functional teams to translate operational requirements into reliable, maintainable software solutions

**Technologies:** C#, .NET, Blazor, Python, Power BI, Machine Learning

## **Software Developer — National Institute of Public Administration (Contract)**

**Oct 2020 – Apr 2021**

- Implemented and enhanced application features to improve system functionality, usability, and performance
- Designed user-friendly software interfaces that simplified administrative workflows and reduced manual effort
- Introduced automation solutions using Python to reduce repetitive tasks and improve operational efficiency
- Collaborated with stakeholders to translate requirements into maintainable software solutions
- Worked across a mixed technology stack including PHP, JavaScript, Java, JSP, and Python

## **Software Engineer (Intern) — The Global Air Drone Academy**

**Jul 2020 – Apr 2021**

- Developed Python scripts to automate drone operations and support flight-related workflows
- Assisted in building and testing drone systems, gaining hands-on experience with hardware–software integration

- Trained students on safe drone operation and compliance with aviation safety regulations
- Supported instructional sessions on drone assembly and system configuration
- Contributed to operational documentation and technical support for training programs

## **Information Technology Intern — Zambia Industrial Commercial Bank (ZICB)**

**Aug 2020 – Oct 2020**

- Developed a user request management system using C# and PHP, supporting internal service workflows
- Assisted in bringing Wincor ATM systems into production, supporting testing and deployment activities
- Provided technical support for Flexcube card-link systems, contributing to system reliability
- Supported IT operations and troubleshooting activities within a regulated banking environment
- Gained hands-on experience working with enterprise financial systems and infrastructure

## **Education**

### **The ZCAS University**

**Master's Degree, Data Science** — Jan 2026 – Jun 2027 (in progress)

### **Mulungushi University**

**Bachelor's Degree, Computer Science** — 2016 – 2020

- Graduated with Distinction (Summa Cum Laude)
- Activities and Societies: Dev × Lusaka
- Research Project: *Crop disease detection using Convolutional Neural Networks*

## **Licenses & Certifications**

- **AWS Certified Solutions Architect – Associate** (Amazon Web Services, issued Sep 2025, expires Sep 2028)
- **Certified Kubernetes Administrator (CKA)** (The Linux Foundation, issued Dec 2024)
- **AWS Certified Machine Learning – Specialty** (Amazon Web Services, issued Oct 2023, expires Oct 2026)
- **TensorFlow Developer Certificate in 2023: Zero to Mastery** (Zero To Mastery Academy, issued Nov 2022)
- **Responsive Web Design** (freeCodeCamp, issued Jul 2020)
- **TensorFlow Developer Certificate** (TensorFlow, issued Nov 2022)

- **Huawei Certified Network Associate (HCIA) Routing and Switching** (Huawei, issued Jan 2020)

## **Key Skills**

- Backend & API Development
- Python (FastAPI), Java, C# (.NET, Blazor), PHP
- AWS (S3, Athena, Translate, Kendra, SageMaker)
- RESTful API Design & Integration
- Cloud Architecture & Infrastructure
- Data Processing & Automation
- Power BI & Data Visualization
- CI/CD & Deployment Workflows
- Machine Learning (Foundational)